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3 Search Theory

This section is based on McCall (1970), Chapter 6 from Ljungqvist and Sargent (2012) (LS thereafter), and Chapter 10.7-10.9 from SLP.

Environment

- There is an unemployed worker searching for a job.
- Preferences are linear: $E \sum_{t=0}^{\infty} \beta^t c_t$.
- Each period while unemployed, the worker receives one job offer $w \sim F(w)$, $w \in [0, \bar{w}]$ (no offer can be modeled as an offer of zero).
- Given an offer, he decides whether to accept or reject
 - If accept, receive w forever.
 - If reject, remain unemployed, receive an unemployment benefit of b today and tomorrow start over.
- Assume that $b < \bar{w}$
- Assume there is no lending or borrowing
- Therefore, $c_t = w$ if employed and $c_t = b$ if unemployed (think of w as the after-tax wage, since b is not explicitly funded by a tax-transfer system).
- Assume there is no recall (cannot go to a previous offer) and no on-the-job search.

Analysis

What is the optimal search strategy (stopping rule) for an unemployed worker?

Denote by $v(w)$ the value to an unemployed worker with the wage offer w . It satisfies the following functional equation:

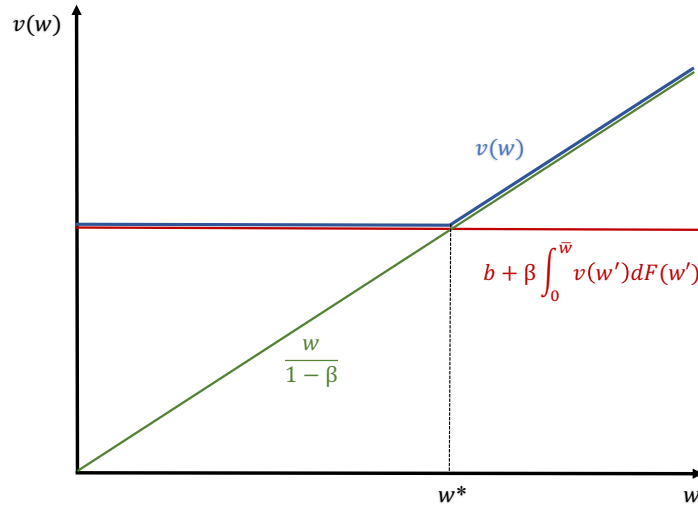
$$v(w) = \max \left\{ \underbrace{\frac{w}{1-\beta}}_{\text{Accept}}, \underbrace{b + \beta \int_0^{\bar{w}} v(w') dF(w')}_{\text{Reject}} \right\}.$$

We can show that the RHS defines an operator that maps the set of continuous bounded functions $f: [0, \bar{w}] \rightarrow \mathbb{R}$ on itself, and it is a contraction with modulus β . Then, there exists a unique fixed point $v(w)$ that satisfies the above Bellman equation.

Define w^* as

$$\frac{w^*}{1-\beta} = b + \beta \int_0^{\bar{w}} v(w') dF(w').$$

Figure 1: Reservation Wage Strategy



The optimal strategy is the “reservation wage strategy”:

- accept if $w \geq w^*$
- reject if $w < w^*$

One simple point: the duration of an unemployment spell for a worker following the optimal strategy is a random variable. What is the expected value of unemployment duration for a worker who begins unemployed today?

Define $p = 1 - F(w^*)$ —the probability to get an offer $w \geq w^*$ and hence become employed. Then the probability of the unemployment spell to be equal to 1 is p , to be equal to 2 is $(1-p)p$, to be equal to 3 is $(1-p)^2p$, ..., to be equal to n is $(1-p)^{n-1}p$. The expected unemployment duration is then

$$\begin{aligned} \sum_{n=1}^{\infty} n(1-p)^{n-1}p &= p \frac{d}{dp} \left[- \sum_{n=1}^{\infty} (1-p)^n \right] = p \frac{d}{dp} \left[- \frac{1}{1-(1-p)} + 1 \right] = p \frac{d}{dp} \left(-\frac{1}{p} \right) \\ &= \frac{p}{p^2} = \frac{1}{p} = \frac{1}{1-F(w^*)}. \end{aligned}$$

If w^* is higher, then $F(w^*)$ is higher, and $1/(1-F(w^*))$ is also higher. Of course, w^* is endogenous.

We will now determine w^* as a function of the primitives of the model (b, β, F) .

General Point for DP

If we know $g(x)$, then we can solve easily for $v(x)$, since

$$v(x) = F(x, g(x)) + \beta F(g(x), g(g(x))) + \beta^2 F(g(g(x)), g(g(g(x)))) + \dots$$

In the context of our model, the optimal decision rule is described by $w^* \Rightarrow$ we can solve for $v(w)$ in terms of w^* . Then, we can substitute for $v(w)$ in the equation

$$\frac{w^*}{1-\beta} = b + \beta \int_0^{\bar{w}} v(w') dF(w')$$

and have an equation that contains only w^* and the model's primitives.

How to express $v(w)$ in terms of w^ :*

From Figure 1, we can write $v(w) = \max\{\frac{w}{1-\beta}, \frac{w^*}{1-\beta}\} \Rightarrow$

$$\begin{aligned}
w^* &= (1-\beta)b + \beta \int_0^{\bar{w}} \max\{w, w^*\} dF(w) \\
&= (1-\beta)b + \beta w^* \int_0^{w^*} dF(w) + \beta \int_{w^*}^{\bar{w}} w dF(w) \\
&= (1-\beta)b + \beta w^* \int_0^{w^*} dF(w) + \beta w^* \int_{w^*}^{\bar{w}} dF(w) - \beta w^* \int_{w^*}^{\bar{w}} dF(w) + \beta \int_{w^*}^{\bar{w}} w dF(w) \\
&= (1-\beta)b + \beta w^* + \beta \int_{w^*}^{\bar{w}} (w - w^*) dF(w) \Rightarrow
\end{aligned}$$

$$\underbrace{w^* - b}_{\substack{\text{MC of additional} \\ \text{search period if have} \\ \text{an offer } w^* \text{ in hand}}} = \underbrace{\frac{\beta}{1-\beta} \int_{w^*}^{\bar{w}} (w - w^*) dF(w)}_{\substack{\text{MB of an additional} \\ \text{search period}}} \quad (1)$$

Agustina's way of writing this equation:

$$w^* = b + \frac{\beta}{1-\beta} \int_{w^*}^{\bar{w}} (w - w^*) dF(w).$$

This expression tells us that the reservation wage w^* exceeds the unemployment benefit b by the expected present discounted value of all future wage offers exceeding w^* . Why exceeding w^* ? Because the worker can secure w^* by remaining unemployed.

For some issues, it is useful to express equation (1) differently. Applying integration by parts,

$$\begin{aligned}
\int_{w^*}^{\bar{w}} (w - w^*) dF(w) &= (w - w^*) F(w) \Big|_{w^*}^{\bar{w}} - \int_{w^*}^{\bar{w}} F(w) dw \\
&= (\bar{w} - w^*) F(\bar{w}) - (w^* - w^*) F(w^*) - \int_{w^*}^{\bar{w}} F(w) dw = \int_{w^*}^{\bar{w}} 1 dw - \int_{w^*}^{\bar{w}} F(w) dw \\
&= \int_{w^*}^{\bar{w}} (1 - F(w)) dw.
\end{aligned}$$

Therefore, we have

$$w^* - b = \frac{\beta}{1-\beta} \int_{w^*}^{\bar{w}} (1 - F(w)) dw. \quad (2)$$

Comparative Statics Exercises

1. What happens to w^* if $b \uparrow$?

Recall $\frac{w^*}{1-\beta} = b + \beta \int_0^{\bar{w}} v(w) dF(w)$. One might be tempted to say directly that $b \uparrow \Rightarrow RHS \uparrow \Rightarrow w^* \uparrow$. But $v(w)$ is endogenous and is affected by b .

Define

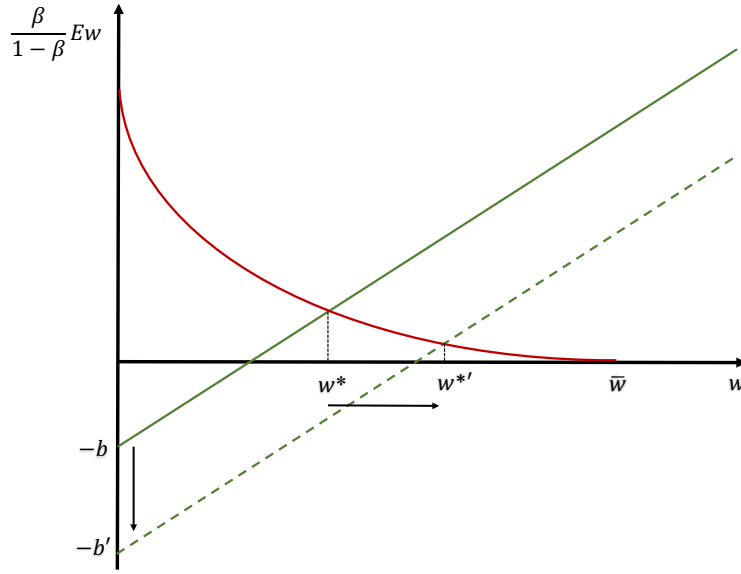
$$h(w) = \frac{\beta}{1-\beta} \int_w^{\bar{w}} (w' - w) dF(w')$$

We have

$$h(0) = \frac{\beta}{1-\beta} Ew, \quad h(\bar{w}) = 0$$

$$h'(w) \underbrace{=}_{\text{verify}} -\frac{\beta}{1-\beta} [1 - F(w)] < 0, \quad h''(w) = \frac{\beta}{1-\beta} F'(w) > 0$$

Figure 2: Increase in b

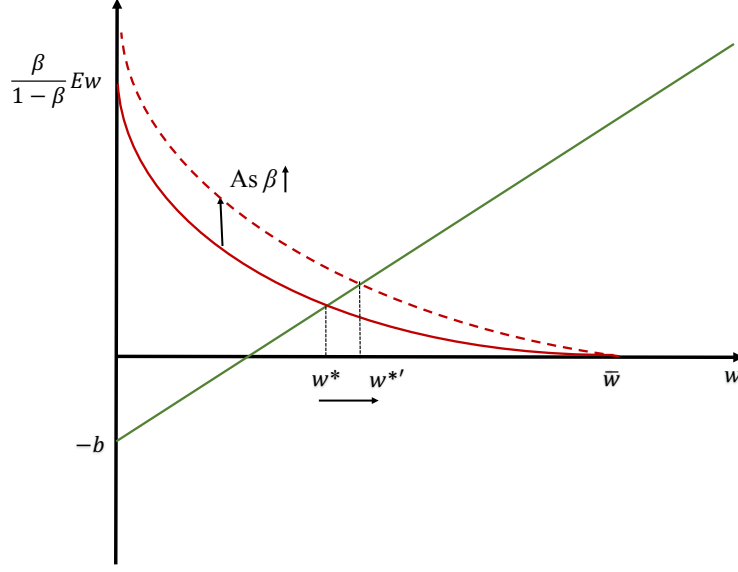


As Figure 2 shows, $b \uparrow \Rightarrow w^* \uparrow$

The intuition behind this is as follows. The marginal cost of search decreases, which makes it less costly to continue waiting for a higher wage. Notice that since the unexpected unemployment duration is increasing in w^* , and thus increasing in b .

2. What happens to w^* if $\beta \uparrow$?

Figure 3: Increase in β



As Figure 3 shows, $\beta \uparrow \Rightarrow w^* \uparrow$

3. *What happens to w^* if there is a mean-preserving increase in spread of the distribution $F(w)$?*

Let $F(w, \theta)$ be a family of distributions on $[0, \bar{w}]$ indexed by $\theta \in [0, 1]$ such that

- (i) Mean is independent of θ : $\int_0^{\bar{w}} w dF(w, \theta)$ is independent of θ .
- (ii) Single-crossing property: $\forall \theta_1, \theta_2 \in [0, 1], \theta_1 > \theta_2, \exists \hat{w} : F(w, \theta_1) > F(w, \theta_2)$ for $0 < w < \hat{w}$, and $F(w, \theta_1) < F(w, \theta_2)$ for $\hat{w} < w < \bar{w}$.

Each $F(w, \theta)$ is associated with a different w^* . Now we ask how w^* and the worker's utility change as $\theta \uparrow$.

Claim 1. *If $F(w, \theta)$ satisfies (i) and (ii), and $\theta_1 > \theta_2$, then*

$$(iii) \int_0^w (F(w', \theta_1) - F(w', \theta_2)) dw' > 0 \quad \forall w \in (0, \bar{w}).$$

Proof: If $0 < w < \hat{w}$, where $\hat{w}$ is the single-crossing point for $F(w, \theta_1)$ and $F(w, \theta_2)$, then $F(w', \theta_1) > F(w', \theta_2) \quad \forall w' \in (0, \hat{w})$ by (ii), so we are done.

Also note that (integrating by parts)

$$\int_0^{\bar{w}} F(w', \theta) dw' = \underbrace{w' F(w', \theta) \Big|_0^{\bar{w}}}_{=\bar{w}} - \int_0^{\bar{w}} w' dF(w', \theta)$$

and both terms are independent of $\theta \Rightarrow$

$$\int_0^{\bar{w}} F(w', \theta_1) dw' = \int_0^{\bar{w}} F(w', \theta_2) dw'.$$

Express

$$\begin{aligned} \int_0^w F(w', \theta) dw' &= \int_0^{\bar{w}} F(w', \theta) dw' - \int_w^{\bar{w}} F(w', \theta) dw' \\ &= \underbrace{\bar{w} - \int_0^{\bar{w}} w' dF(w', \theta)}_{\text{independent of } \theta} - \int_w^{\bar{w}} F(w', \theta) dw'. \end{aligned}$$

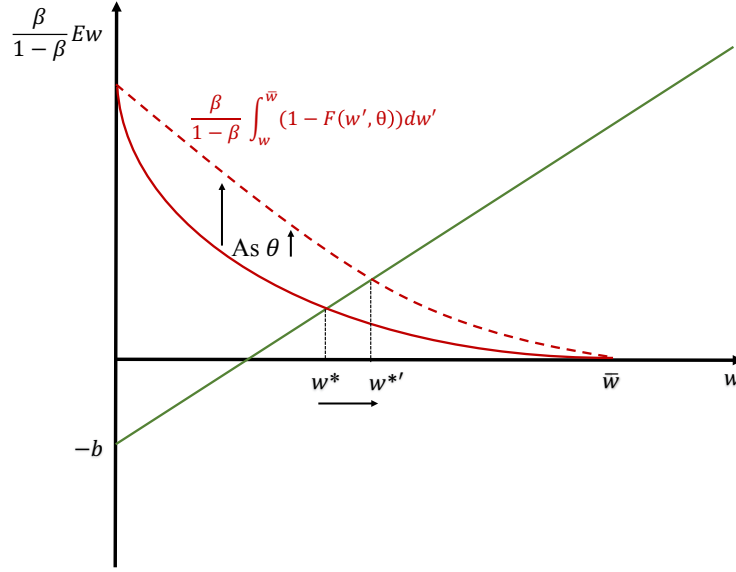
Then, for $w > \hat{w}$

$$\int_0^w [F(w', \theta_1) - F(w', \theta_2)] dw' = \int_{w > \hat{w}}^{\bar{w}} \underbrace{[F(w', \theta_2) - F(w', \theta_1)]}_{>0 \text{ for } w' > \hat{w} \text{ by (ii)}} dw' > 0.$$

To see how an increase in θ changes w^* , use equation (2).

Claim 1 implies that $\int_0^w F(w', \theta) dw'$ is increasing in $\theta \forall w$. Since $\int_0^{\bar{w}} F(w', \theta) dw'$ is constant, $\int_w^{\bar{w}} F(w', \theta) dw'$ is decreasing in $\theta \forall w$. This implies that $\int_w^{\bar{w}} (1 - F(w', \theta)) dw'$ is increasing in θ .

Figure 4: Increase in θ

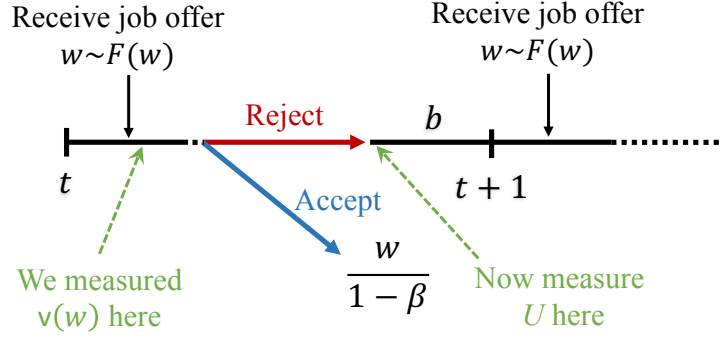


Hence, we can conclude that an increase in θ increases w^* . Under a mean-preserving increase in risk, the higher incidence of very good wage offers increases the value of searching for a job.

Recall $v(w) = \max\{\frac{w}{1-\beta}, \frac{w^*}{1-\beta}\}$. Hence $v(w)$ is also increasing in θ . The unemployed worker is better off with more risk (higher variance in wage offers).

An Alternative Recursive Formulation

Figure 5: An Alternative Formulation with U

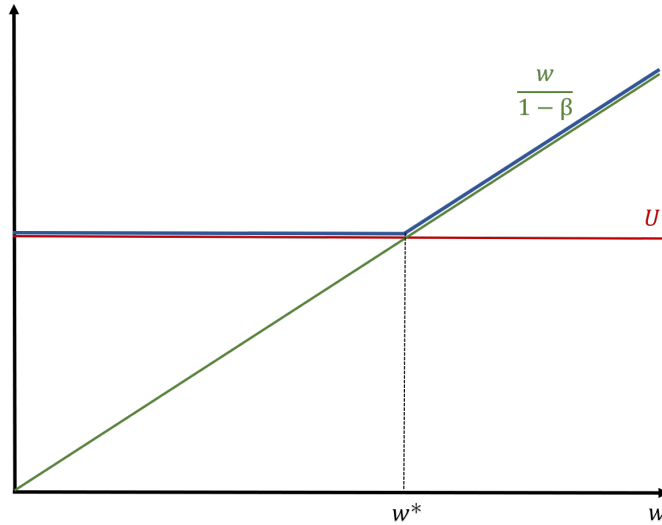


Let U be the best that an agent can do given that he is unemployed and he behaves optimally.

$$U = b + \beta \int_0^{\bar{w}} \max \left\{ \frac{w}{1-\beta}, U \right\} dF(w)$$

Using our previous results we can show that this equation has a unique solution U , as shown in Figure 6.

Figure 6: Alternative – Reservation Wage Strategy



The reservation wage has to be such that $\frac{w^*}{1-\beta} = U$. Substituting in the Bellman

equation,

$$\begin{aligned}\frac{w^*}{1-\beta} &= b + \beta \int_0^{\bar{w}} \max \left\{ \frac{w}{1-\beta}, \frac{w^*}{1-\beta} \right\} dF(w) \Rightarrow \\ w^* &= (1-\beta)b + \beta \int_0^{\bar{w}} \max\{w, w^*\} dF(w)\end{aligned}$$

Using the same steps as before,

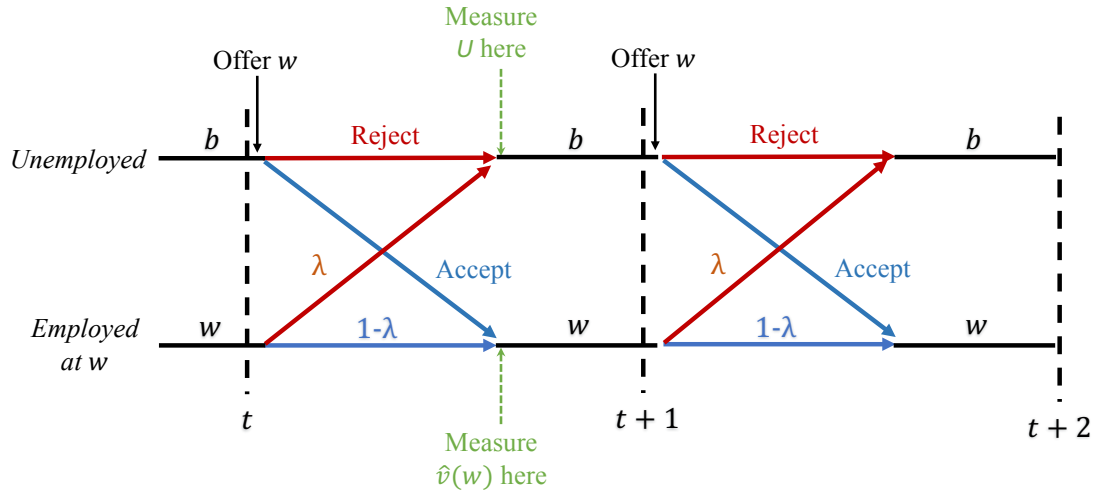
$$w^* - b = \frac{\beta}{1-\beta} \int_{w^*}^{\bar{w}} (w - w^*) dF(w)$$

Two Extensions of the Basic Search Model

1. *Jobs Don't Last Forever*

Consider the same environment as before, except now an employed worker loses his job with probability λ at the beginning of each period (layoff).

Figure 7: Timing with Layoffs



Let U be the value to the unemployed worker and $\hat{v}(w)$ the value to a worker who is employed with wage w . Figure 7 shows the timing in this case. The expressions for U and $\hat{v}(w)$ are as follows.

$$\hat{v}(w) = w + \beta[\lambda U + (1-\lambda)\hat{v}(w)] \quad (3)$$

$$U = b + \beta \int_0^{\bar{w}} \max\{\hat{v}(w), U\} dF(w) \quad (4)$$

Using our standard methods, we can show that these equations have unique solutions for U and $\hat{v}(w)$.

In order to show that a reservation wage is the optimal strategy, it is sufficient to show that $\hat{v}(w)$ is increasing in w and that $\hat{v}(0) < U$. From equation (3)

$$\begin{aligned} [1 - \beta(1 - \lambda)]\hat{v}(w) &= w + \beta\lambda U \Rightarrow \\ \hat{v}(w) &= \frac{w + \beta\lambda U}{1 - \beta(1 - \lambda)}, \end{aligned} \quad (5)$$

where $\hat{v}(w)$ is increasing in w since $1 - \beta(1 - \lambda) > 0$ ($1 > \beta \geq \beta(1 - \lambda)$). Moreover,

$$\hat{v}(0) = \frac{\beta\lambda U}{1 - \beta(1 - \lambda)} < U \Leftrightarrow \beta\lambda < 1 - \beta + \beta\lambda \Leftrightarrow \beta < 1.$$

We want to express w^* in terms of the primitives. We know that w^* is such that $\hat{v}(w^*) = U$. From (3) we have

$$\begin{aligned} U &= w^* + \beta[\lambda U + (1 - \lambda)U] \Rightarrow \\ U &= \frac{w^*}{1 - \beta} \end{aligned} \quad (6)$$

Equation (6) states that the value of unemployment has to be equal to working at the reservation wage forever, which makes you indifferent between working at w^* and not working at all (this also explains why there is no λ in (6)).

From equation (5) we have

$$\hat{v}(w) = \frac{w + \beta\lambda \frac{w^*}{1 - \beta}}{1 - \beta(1 - \lambda)} = \frac{1}{1 - \beta} \left(\frac{(1 - \beta)w + \beta\lambda w^*}{1 - \beta(1 - \lambda)} \right). \quad (7)$$

Substituting (6) and (7) into (4),

$$\frac{w^*}{1 - \beta} = b + \beta \int_0^{\bar{w}} \max \left\{ \frac{1}{1 - \beta} \frac{(1 - \beta)w + \beta\lambda w^*}{1 - \beta(1 - \lambda)}, \frac{w^*}{1 - \beta} \right\} dF(w).$$

Note:

$$\begin{aligned} \frac{(1-\beta)w + \beta\lambda w^*}{1-\beta(1-\lambda)} > w^* &\iff (1-\beta)w + \beta\lambda w^* > (1-\beta(1-\lambda))w^* \iff \\ (1-\beta)w &> (1-\beta)w^* \iff w > w^*. \end{aligned}$$

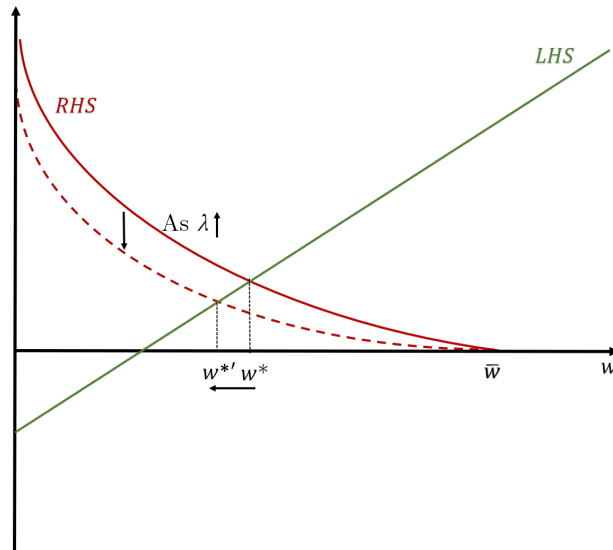
Thus

$$\begin{aligned} \frac{w^*}{1-\beta} &= b + \frac{\beta}{1-\beta} \int_0^{w^*} w^* dF(w) + \frac{\beta}{1-\beta} \int_{w^*}^{\bar{w}} \frac{(1-\beta)w + \beta\lambda w^*}{1-\beta(1-\lambda)} dF(w) \xRightarrow{\text{Verify}} \\ w^* - b &= \frac{\beta}{1-\beta(1-\lambda)} \int_{w^*}^{\bar{w}} (w - w^*) dF(w). \end{aligned}$$

When $\lambda = 0$ the problem is the same as before.

All the previous comparative statics go through. Figure shows what happen when the probability of layoff increases: $\lambda \uparrow \Rightarrow 1-\lambda \downarrow \Rightarrow 1-\beta(1-\lambda) \uparrow \Rightarrow \frac{\beta}{1-\beta(1-\lambda)} \downarrow \Rightarrow w^* \downarrow$. a higher layoff probability makes waiting for a higher wage job less attractive (don't get to keep that job forever).

Figure 8: CS: Increase in λ



Economy with many i.i.d. workers

Let $p = 1 - F(w^*)$ be the probability of going from unemployment to employment.

Let u_t be the fraction of workers unemployed in period t . Then

$$u_{t+1} = \underbrace{(1-p)u_t}_{\text{Unemployed at } t \text{ and don't find a job}} + \underbrace{\lambda(1-u_t)}_{\text{Employed at } t \text{ and lose the job}}.$$

This is a first-order difference equation.

Rewrite $u_{t+1} = (1-p-\lambda)u_t + \lambda$.

There exists a unique steady state: $u^* = (1-p-\lambda)u^* + \lambda \Rightarrow u^* = \frac{\lambda}{p+\lambda}$. The steady-state u^* is increasing in λ and decreasing in p . Convergence to the steady state depends on the model parameters:

- $p + \lambda < 1$: monotone convergence
- $p + \lambda > 1$: oscillating convergence
- $p + \lambda = 1$: $u^* = \lambda$, $u_{t+1} = 0 \cdot u_t + \lambda$, which implies an immediate jump to the steady state.

2. Endogenous Search Intensity

Let $C(a)$ be disutility of search effort required to receive an offer with probability a . C is C^2 , strictly increasing, strictly convex, $C(0) = 0$, $C'(0) = 0$, $\lim_{a \rightarrow 1} C'(a) = \infty$. Assume there are no layoffs.

The value to an unemployed worker is

$$U = \max_a \left\{ b - C(a) + \beta \left[(1-a)U + a \int_0^{\bar{w}} \max \left\{ \frac{w'}{1-\beta}, U \right\} dF(w') \right] \right\} \quad (8)$$

This equation has a unique solution for U .

The optimal behavior is characterized by two rules

1. How hard to search.
2. Which job to accept $\rightarrow$ reservation wage w^*

The optimal search intensity is a constant, a^* .

Characterizing the optimal decision as a function of the primitives

We already know that $U = \frac{w^*}{1-\beta}$. Hence

$$\frac{w^*}{1-\beta} = b - C(a^*) + \beta \left[(1-a^*) \frac{w^*}{1-\beta} + a^* \int_0^{\bar{w}} \max \left\{ \frac{w'}{1-\beta}, \frac{w^*}{1-\beta} \right\} dF(w') \right]$$

(Note that if $a^* = 1$, this is essentially the same as the original model except the unemployment benefit is $b - C(1)$).

Doing the usual manipulations this equation reduces to (verify):

$$w^* = \underbrace{b - C(a^*)}_{\substack{\text{flow utility of} \\ \text{of searching} \\ \text{one period}}} + \underbrace{\frac{a^* \beta}{1-\beta} \int_{w^*}^{\bar{w}} (w - w^*) dF(w)}_{\substack{\text{expected return of searching} \\ \text{one more period} \\ \text{(marginal benefit)}}}, \quad (9)$$

where $w^* - b + C(a^*)$ is the marginal cost.

Differentiating (8) w.r.t. a gives the optimal choice of a given w^* .

$$-C'(a) - \beta U + \beta \int_0^{\bar{w}} \max \left\{ \frac{w'}{1-\beta}, U \right\} dF(w') = 0.$$

Rearranging terms and evaluating at a^*, w^*

$$\begin{aligned} C'(a^*) &= \frac{\beta}{1-\beta} \left[\int_0^{\bar{w}} \max\{w', w^*\} dF(w) - w^* \right] \\ \underbrace{C'(a^*)}_{\substack{\text{Marginal cost} \\ \text{of searching} \\ \text{harder}}} &= \underbrace{\frac{\beta}{1-\beta} \int_{w^*}^{\bar{w}} (w' - w^*) dF(w')}_{\substack{\text{Marginal benefit of an} \\ \text{increased probability of} \\ \text{getting a higher wage offer}}} \end{aligned} \quad (10)$$

Equations (8) and (9) give us two equations and two unknowns, w^* and a^* , and the model primitives. Substitute (10) into (9)

$$w^* = b - C(a^*) + a^* C'(a^*). \quad (11)$$

Diagrammatic representation of the optimal choice.

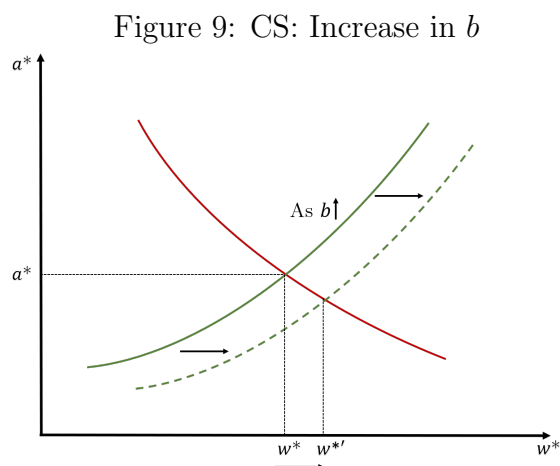
In (10) the left-hand side is increasing in a^* and the right-hand side is decreasing in $w^* \Rightarrow$ (10) gives a downward-sloping curve in the (w^*, a^*) space. Think of the curve $a^*(w^*)$ given by (10) as defining the optimal choice a^* given w^* .

We cannot do the same with (9) (first term is $\downarrow$ in a^* and the second $\uparrow$ in a^*). Therefore, we need to use (11).

The left-hand side is increasing in w^* , the right-hand side is increasing in a^* since

$$\frac{\partial}{\partial a}[-C(a) + aC''(a)] = -C'(a) + C'(a) + aC'''(a) = aC'''(a) > 0$$

Hence, (11) gives an upward-sloping curve in the (w^*, a^*) space. Think of the curve $w^*(a^*)$ given by (11) as defining the optimal choice w^* given a^* .



Suppose that $b \uparrow$. Nothing happens to (10), but (9) shifts to the right $\Rightarrow a^* \downarrow$ and $w^* \uparrow$. This is an example of moral-hazard problems associated with unemployment insurance.

References

- LJUNGQVIST, L. AND T. J. SARGENT (2012): *Recursive Macroeconomic Theory*, MIT Press.
- MCCALL, J. J. (1970): “Economics of Information and Job Search,” *The Quarterly Journal of Economics*, 84, 113–126.